Individual Self-Study Assignment and Submission Template

Leveling Module 1: Applied ML Workflow Mini-Project

CS640: Applied Artificial Intelligence and Machine Learning

Field	Student Response
Student Name	
Date	
Dataset / Task Name	
Notebook / Repository Link	

1. Assignment Overview

Purpose. This individual mini-project gives you hands-on practice with the applied machine learning workflow before the class moves deeper into the larger CS640 team capstone project. The goal is not to build the most advanced model. The goal is to demonstrate a disciplined workflow: define a task, build a baseline, validate appropriately, choose metrics, inspect errors, document experiments, and possibly connect what you learn to the larger capstone.

GUIDANCE Expected scope. Keep this mini-project small. A successful submission may use a compact public dataset. The written response should normally be 2-4 pages, plus code/notebook evidence as an appendix or link.

2. Two Tracks in CS640

Track	Purpose	Output
Individual leveling mini-project	Practice the ML workflow personally using a small, controlled task.	This completed worksheet plus code/notebook evidence.
Team capstone project	Apply the workflow to a larger applied AI/ML system across the semester.	Team proposal, baseline, prototype, final demo, and final report.

Connection. The mini-project is where you practice the workflow. The capstone is where your team applies the workflow at larger scale. Note: Connection to the capstone is optional, not a requirement.

WORKFLOW AND PROJECT FRAMING

From task selection to a testable plan

Keep the task small enough to complete, inspect, and explain.

3. Individual Mini-Project Instructions

Complete the following workflow:

- Select a small dataset, corpus, or ML task.
- Define the prediction, classification, regression, retrieval, or inference task.
- Identify input features and the target/output.
- Build a simple baseline model or baseline system.
- Choose a validation strategy and primary metric.
- Try one improvement: a model change, feature change, preprocessing change, threshold change, or representation change.
- Inspect selected errors or edge cases.
- Document your modeling iterations in an experiment log.
- Write a short paragraph explaining how this workflow informs the larger team capstone.

4. Dataset and Task Description

Prompt	Student Response
Dataset source or URL	
Dataset/corpus context: Who collected it? What does it represent?	
Number of instances/samples/documents	
Input features or inputs	
Target/output variable	
Task type: classification, regression, text classification, retrieval, recommendation, other	
Operational context: Who might use this system and for what decision or output?	
Known dataset issues: missing values, imbalance, noise, bias, duplicates, small sample size, etc.	

BASELINE AND VALIDATION

Establish a credible starting point

Choose a baseline, evaluation design, and metrics before optimizing.

5. Baseline Taxonomy

A baseline does not need to be impressive. It needs to be clear, reproducible, and beatable. Use the table below to select a reasonable baseline for your project type.

Project Type	Reasonable Baseline
Classification	Majority class, logistic regression, decision tree
Regression	Mean predictor, linear regression
Text classification	Bag-of-words or TF-IDF + logistic regression
RAG / document question answering	Keyword search + manual answer or simple retrieval-only baseline
Computer vision	Simple pretrained feature extractor + classifier
Reinforcement learning	Random policy or simple heuristic policy
Recommendation	Popularity-based recommender

6. Baseline and Validation Plan

Prompt	Student Response
Baseline model/system selected	
Why this is an appropriate baseline	
Validation strategy: train/test split, validation set, cross-validation, temporal split, etc.	
Primary metric	
Why this metric is appropriate	
Secondary metric(s), if any	
What would count as meaningful improvement over baseline?	

ITERATION

Test one meaningful improvement

Change one element, state the hypothesis, and compare it with the baseline.

7. Improvement Attempt

ITERATION RULE Record what changed, why you expected it to help, what happened, and whether the evidence supports keeping the change.

Prompt	Student Response
Improvement attempted: model change, feature change, preprocessing change, threshold change, representation change, etc.	
Hypothesis: Why did you expect this change to help?	
Result compared to baseline	
Decision: keep, reject, or investigate further	

EVIDENCE STANDARD A change is useful only when its result is evaluated under the same validation strategy and metric used for the baseline.

DATA INTEGRITY

Leakage checklist

Check whether information crossed a boundary it should not have crossed.

8. Leakage Checklist

Instruction. Answer Yes, No, or N/A for each item and briefly explain any risk. Leakage can produce impressive metrics that do not survive contact with reality.

Checklist Item	Yes / No / N/A	Brief Explanation
Did I split the data before fitting preprocessing steps such as scaling, encoding, feature selection, or PCA?		
Are duplicate or near-duplicate records present across train/test or validation sets?		
Does any feature contain future information that would not be available at prediction time?		
Is the target indirectly encoded in one or more features?		
For time-based data, did I preserve chronological order when appropriate?		
Did I avoid repeatedly tuning decisions on the final test set?		
For document-based projects, could related chunks from the same source appear in both development and evaluation sets?		
Did I document any unresolved leakage concern?		

DOCUMENT THE RISK If a concern cannot be resolved within the mini-project, state it explicitly and explain how it limits the result.

MODEL BEHAVIOR

Error analysis

Aggregate metrics show how much the model failed; selected cases help explain why.

9. Error Analysis

Instruction. Do not document every test case. Evaluate the full validation/test set using aggregate metrics, then select 5-10 interesting cases for manual inspection. Focus on mistakes, edge cases, high-confidence wrong predictions, large regression errors, underperforming classes, or examples that reveal model limitations.

Case ID	Actual Value/Class	Predicted Value/Class	Error Type or Error Size	Possible Cause	Possible Fix

Summary of failure pattern. In 3-5 sentences, explain the main pattern you observed in the selected errors:

LEARNING ACROSS ITERATIONS

Experiment log and capstone connection

Document the modeling path, then carry the lesson into the team project.

10. Experiment Log

Instruction. The experiment log documents your modeling thought process. Minimum expectation: one baseline experiment, one improvement experiment, and one conclusion about what worked or failed.

Experiment	Change Made	Hypothesis	Result	Decision / Lesson
E1	Baseline model/system	Establish starting point		
E2				
E3				

11. Team Capstone Connection

Write one paragraph connecting this mini-project to the larger team capstone. Address at least two of the following:

- What kind of baseline might the team capstone need?
- What metric or evaluation issue might apply to the capstone?
- What leakage risk might appear in the capstone?
- What error-analysis strategy might transfer to the capstone?
- What did this mini-project teach you about project feasibility?
- What role might you be able to play in the team capstone based on this experience?

COMPLETION STANDARD

Submission expectations and rubric

Submit the evidence of your workflow and use the rubric as a final quality check.

12. Submission Expectations

Submit the following:

- Completed worksheet as a Word document or PDF.
- Code/notebook file or repository link, if applicable.
- Any dataset link or short note explaining where the dataset came from.
- Evidence of baseline and improved-model results.
- Completed leakage checklist, error analysis table and experiment log.

Rubric

Component	Points
Task and dataset description	10
Baseline model/system	15
Validation strategy and metric choice	15
Improved model or feature/system change	15
Leakage checklist	10
Error analysis	20
Experiment log	15
Total	100

TOTAL The rubric is worth 100 points. Strong work is reproducible, appropriately validated and transparent about leakage and errors.